Exploring Limitations and Benefits of AI-Generation Dectection Paradigms

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Abstract

With the advent of image generation models, the authenticity of images–whether they be real images or art–has come into great controversy. In particular, the detection of images to be AI generated or human-created has become an increasingly important task. Currently, while multiple approaches have been taken to improve the accuracy of AI image detection, such as through a variety of algorithms, models, and even manipulations of the input images themselves, there are still limitations on the efficacy of such approaches in the inability to generalize or accuracy. Datasets used will be the AI Recognition Dataset, a collection of 17.9k AI-Generated images from models such as DALL-E and Midjourney, and a collection of 4k human-painted images from the internet and the Art Dataset, a collection of 500k art/paintings by genre. A random subset can be used to address imbalance in real versus AI images from the first dataset.

1. Introduction

The detection of AI-generated images has attracted significant research attention across multiple paradigms, each targeting a different class of artifacts or representational cues. Our work draws from and builds upon three major lines of inquiry: frequency-domain forensics, vision-language and multimodal detection, and statistical/entropy-based approaches.

1.1. Vision-Language and Multimodal Detection

Purely visual detectors tend to overfit to generator-specific artifacts, degrading severely on unseen generative models [8, 12]. Vision-language models (VLMs) offer a natural remedy: their joint image–text embedding spaces encode semantic information that is stable across generators, making them better suited for OOD detection. GenDet [14] performs real-versus-fake discrimination directly in the CLIP embedding space, while LASTED [13] uses language-guided contrastive learning to anchor visual features to textual descriptions of real-image properties.

Our multimodal track compares four configurations: (a) a fine-tuned ViT binary classifier; (b) zero-shot CLIP with multi-prompt averaging over five synonymous prompt pairs per class [11]; (c) a ViT encoder fine-tuned to project images into the CLIP embedding space via a learned projection head; and (d) a logit-averaging ensemble of (a) and (c).

1.2. Texture and Entropy-Based Analysis

Statistical analysis has long been used in image forensics to distinguish natural images from synthetic or manipulated content. Entropy-based measures quantify the randomness and complexity of pixel distributions, where Shannon entropy captures global intensity randomness and local entropy measures spatial texture variations. Natural photographs typically exhibit diverse textures due to real-world structures, illumination changes, and sensor noise, whereas AI-generated images often produce smoother textures and more uniform pixel distributions [1, 10].

Frequency-domain analysis has also been widely used for detecting synthetic images. Prior work shows that generative models often fail to reproduce the spectral statistics of natural images, leading to irregularities in the Fourier spectrum and abnormal energy distributions in high-frequency bands. Such spectral artifacts can act as fingerprints for detecting GAN-generated images [3, 4, 6].

Sharpness-based metrics provide another useful signal. Measures such as Laplacian variance and gradient statistics quantify the strength and variability of edge structures in an image. Natural images typically contain stronger and more diverse high-frequency details, while generative models may introduce over-smoothing that reduces edge variability [9]. In addition, simple color distribution statistics, such as the mean and standard deviation of RGB channels, can capture abnormal saturation patterns or unusually uniform color distributions in synthetic images.

Despite these advances, many existing approaches rely on a single feature family or are designed specifically for

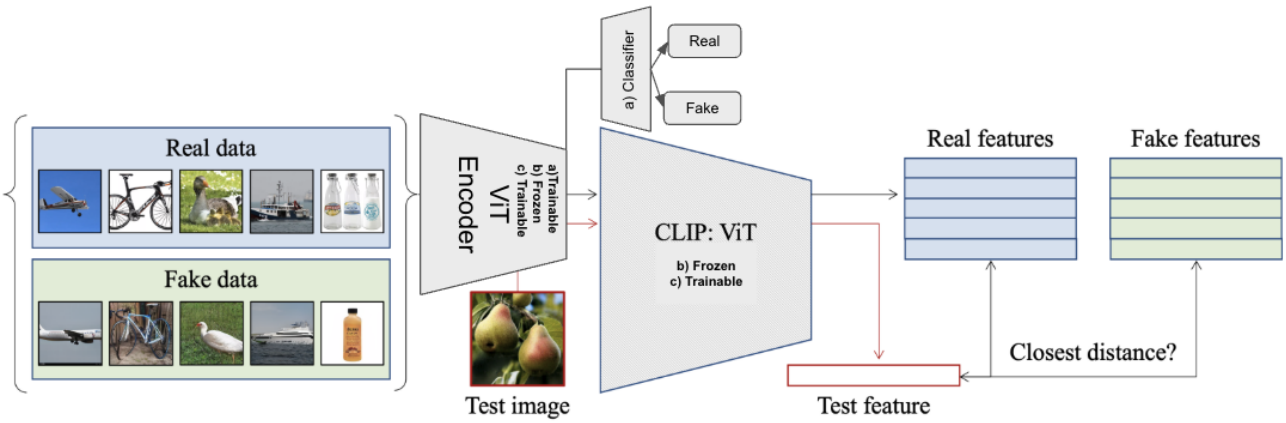


Figure 1. Overview of the Encoder+CLIP pipeline for AI image detection. Training data (real and fake images) is passed through a ViT encoder, which can be operated in three modes: (a) trainable end-to-end with a supervised classifier head, (b) frozen with only the projection head trained, or (c) fully fine-tuned with CLIP alignment. The encoder projects images into the CLIP:ViT embedding space, producing class prototype feature stores for real and fake images respectively. At inference time, a test image is encoded into the same space and classified by nearest-prototype distance. This design allows the model to leverage both supervised visual artifact learning and the semantic generalization of CLIP's pre-trained embedding space.

GAN artifacts. As generative models continue to evolve, there is a growing need for compact and computationally efficient feature representations that integrate multiple statistical cues.

1.3. Frequency-Domain Spectral Analysis

A parallel line of research focuses on the "spectral fingerprints" left by generative architectures. Durall et al. [FIX-HERE] first demonstrated that the up-convolution operations used in nearly all GAN-based generators fail to accurately reproduce the spectral distribution of natural images. While natural images typically exhibit a monotonic energy decay toward high frequencies, AI-generated images often display characteristic artifacts—such as high-frequency energy spikes or a reduced rate of falloff—due to periodic zero-padding and filtering in the upsampling process.

Erukude et al. [5] proposed the use of a ResNet-50 architecture trained directly on the 2D log-magnitude spectral images. A shortcoming addressed was that even models operating on the 2D frequency spectrum may may fail to generalize.

Our work extends these ideas by transitioning from 2D spatial spectral analysis to a 1D representation via Azimuthal Integration. By collapsing the 2D Fourier spectrum into a 1D radially averaged power spectrum, we explicitly model the energy-falloff discrepancy as a function of frequency radius. This approach offers two key advantages. First, it provides a rotation-invariant representation that focuses solely on the rate of spectral decay. Second, it enables the deployment of classical, mathematically grounded classifiers—specifically SVMs with RBF kernels and AdaBoost—which offer transparent insight into classification decisions and computational efficiency than deep CNN baselines.

2. Methods

Our detection framework comprises three independent analysis tracks-multi-modal vision-language classification, frequency-domain spectral analysis, and statistical entropy-based forensics- each targeting a distinct class of artifact. We describe the dataset, the failure-case benchmark construction, and then each track in turn.

2.1. Dataset and Preprocessing

We train and evaluate on the **AI Recognition Dataset**, which contains 17,900 AI-generated images produced by models including DALL-E and Midjourney, paired with approximately 4,000 human-painted images collected from the internet. The dataset presents a natural class imbalance of roughly 4.5:1 (fake:real); we address this through stratified splitting rather than oversampling, preserving the natural distribution during evaluation.

All images are resized to 256×256 pixels prior to feature extraction. For deep vision-based pipelines, images are additionally normalized to the ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). For the statistical forensics track, images are converted to grayscale after channel-level feature extraction, and pixel values are normalized to the range $[0, 1]$.

2.2. Failure-Case Benchmark Construction

A key methodological contribution of this work is the construction of a **Combined Failure Cases Dataset** for stress-testing detector robustness. Each trained model is run on the full held-out test set, and every image that is misclassified by at least one model is retained. The resulting dataset pools 3,611 hard samples-2,578 Fake→Real misclassifications and 1,033 Real→Fake misclassifications-drawn from across all methods (ViT, Zero-Shot CLIP, Encoder+CLIP, Spectral Analysis, and Forensic Approaches).

This benchmark serves two purposes. First, it allows direct measurement of how much each model's performance degrades on genuinely difficult inputs. Second, by annotating each failure with its source model and error type, we can distinguish *visual artifact failures*-cases where the model misses low-level generative cues-from *semantic alignment failures*-cases where the model's high-level representation is insufficiently discriminative.

2.3. Track A: Multimodal Vision-Language Classification

2.3.1. Standard ViT Pipeline (Supervised, Image-Only)

Our supervised baseline is a Vision Transformer (ViT-B/16) [2] pre-trained on ImageNet and fine-tuned end-to-end for binary classification. A linear classification head replaces the original output layer, producing logits for the *Real* and *Fake* classes. The model is trained with a class weighted cross-entropy loss, AdamW optimizer, and a cosine learning rate schedule with linear warmup. This pipeline serves as the performance ceiling for purely visual detection.

2.3.2. Zero-Shot CLIP with Multi-Prompt Averaging

We evaluate zero-shot generalization using CLIP (ViT-B/32) [11], which encodes both images and text in a shared embedding space without any task-specific fine-tuning. Classification is performed by computing the cosine similarity between the image embedding and the text embeddings of two class descriptors, assigning the image to the higher-similarity class.

To mitigate sensitivity to prompt wording, we employ **multi-prompt averaging**: for each class, we define five semantically equivalent prompt templates (e.g., *a real photograph*, *a genuine human artwork*, *an authentic image*) and average the resulting text embeddings before scoring. This strategy, inspired by the prompt ensembling of [11], reduces variance due to incidental lexical choices and provides a cleaner measure of CLIP's intrinsic discriminative capability.

2.3.3. Encoder + CLIP Pipeline (Trained Projection Head)

To combine the generalization benefits of CLIP's semantic space with supervision from labeled data, we introduce a hybrid architecture in which a ViT encoder is trained to project images into the CLIP embedding space via a learned projection head. Formally, let f_θ denote the ViT encoder parameterized by θ, P_ϕ a two-layer MLP projection head, and $\mathbf{c}_{\text{real}}$, $\mathbf{c}_{\text{fake}}$ the (frozen) CLIP text embeddings of the class prompts. The classification score is:

$$\hat{y} = \arg\max_{k \in \{\text{real, fake}\}} \cos\big(P_\phi(f_\theta(\mathbf{x})),\ \mathbf{c}_k\big). \tag{1}$$

We explore two variants: (b) **frozen** CLIP backbone with only P_ϕ trained, and (c) **trainable** CLIP backbone where the entire model is fine-tuned end-to-end. Both variants use contrastive alignment loss combined with cross-entropy supervision.

2.3.4. Ensemble

Our ensemble (variant d) averages the output logits of the fine-tuned ViT (variant a) and the Encoder+CLIP model (variant c):

$$\mathbf{l}_{\text{ens}} = \frac{\mathbf{l}_{\text{ViT}} + \mathbf{l}_{\text{EC}}}{2}. \tag{2}$$

When both models agree the average reinforces the shared prediction; when they disagree, the higher-confidence model's logits dominate. This simple averaging scheme leverages the complementary inductive biases of pure visual classification and vision-language alignment without requiring additional training.

2.4. Track B: Frequency-Domain Spectral Analysis

2.4.1. Motivation

GANs and diffusion models introduce characteristic high-frequency artifacts due to periodic upsampling [4]. While the Fourier spectrum of real images decays monotonically, AI-generated images exhibit unnatural high-frequency energy retention or increase.

2.4.2. Fourier Preprocessing Block

Images undergo four preprocessing steps:

1. **2D FFT**: Spatial image $I \in \mathbb{R}^{H \times W}$ is transformed to its complex Fourier representation $\mathcal{F}(I)$.
2. **FFT Shift**: The zero-frequency component is centered.
3. **Log-Magnitude Scaling**: $M = \log(1 + |\mathcal{F}(I)|)$ compresses the dynamic range to highlight high-frequency artifacts.
4. **Azimuthal Integration**: The 1D radial profile, $P(r)$, is generated by averaging log-magnitudes of Fourier coefficients within each discrete radial bin $r = \sqrt{(x - x_c)^2 + (y - y_c)^2}$:

$$a[r] = \frac{1}{|\mathcal{N}_r|} \sum_{(u,v) \in \mathcal{N}_r} M(u, v)$$

2.4.3. Baseline: ResNet-50 on Full Spectral Images

Following Erukude et al. [5], our deep-network baseline is a ResNet-50 trained directly on the full 2D log-magnitude spectral images $M \in \mathbb{R}^{256\times256}$.

2.4.4. Using Classifiers on Azimuthal Integration Data

To rival the ResNet-50 baseline using computationally shallower models, we train and tune two classical classifiers on the 128-dimensional Azimuthal power spectrum vectors $\mathbf{a}$:

- **SVM with RBF Kernel**: A soft-margin SVM maps $\mathbf{a}$ to a high-dimensional feature space for non-linear separation. A wider soft margin improves generalization over deeper networks that may overfit [5]. We tuned the regularization parameter $C \in \{0.01, 0.10, 1.00, 10.00, 100.00\}$ using balanced class weights to optimize the margin-error trade-off. To validate Azimuthal Integration as an effective dimensionality reduction technique, we also compared it against a baseline SVM trained on raw 2D Fourier spectra flattened and reduced via PCA (configured to 95% variance).
- **AdaBoost**: An ensemble of weak learners [7]. AdaBoost inherently provides feature importance scores across the 128 frequency bins, offering direct interpretability. We tuned the number of estimators ($n \in \{10, 50, 100, 200, 300\}$) and evaluated three weak learner architectures (Depth-1 Stumps, Depth-3 Trees, and Logistic Regression) to optimize ensemble robustness.

2.5. Track C: Statistical Entropy-Based Forensics

2.5.1. Motivation

AI-generated images often exhibit distinct statistical artifacts, such as excessively smooth textures and unusually uniform pixel distributions. Unlike natural photographs, which contain inherent randomness due to real-world imaging processes, synthetic images tend to lack this variability. These deviations from natural image statistics can be captured using a compact set of carefully designed statistical features, enabling reliable detection without relying on deep neural networks.

2.5.2. Feature Extraction

We extract an 11-dimensional feature vector from each image, organized into three groups that capture complementary statistical characteristics of the image.

Entropy Features (3 dimensions).

- **Global (Shannon) entropy**: measures the uncertainty or randomness in the pixel intensity distribution of an image, providing a quantitative estimate of its information content.

$$H = -\sum_i p_i \log p_i \tag{3}$$

where p_i is the probability of intensity level i in the grayscale pixel histogram, measuring overall pixel randomness.

- **Local entropy**: the mean Shannon entropy computed over non-overlapping 8×8 patches:

$$H_{\text{local}} = \frac{1}{N}\sum_{k=1}^{N}\left(-\sum_i p_{i,k}\log p_{i,k}\right) \tag{4}$$

where $p_{i,k}$ denotes the probability of intensity i in patch k and N is the number of patches.

- **Gradient entropy**: Shannon entropy computed over the histogram of gradient magnitudes obtained from the Sobel operator:

$$G = \sqrt{G_x^2 + G_y^2}, \qquad H_{\text{grad}} = -\sum_j q_j \log q_j \tag{5}$$

where G_x and G_y are Sobel gradients and q_j represents the probability distribution of gradient magnitudes.

Sharpness and Frequency Features (2 dimensions).

- **Laplacian variance**: quantifies the spread of Laplacian responses in an image, where higher values indicate stronger edges and finer texture details, defined as

$$L = \nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2} \tag{6}$$

$$\sigma_L^2 = \frac{1}{MN}\sum_{x,y}(L(x,y) - \mu_L)^2 \tag{7}$$

where I is the grayscale image, $L(x, y)$ is the Laplacian response, and μ_L is its mean. Larger variance indicates sharper images.

- **High-frequency energy ratio**: computed in the frequency domain using the 2D Fourier transform:

$$F(u, v) = \mathcal{F}\{I(x, y)\} \tag{8}$$

$$R_{\text{HF}} = \frac{\sum_{(u,v)\in\Omega_{\text{HF}}} |F(u,v)|^2}{\sum_{(u,v)} |F(u,v)|^2} \tag{9}$$

where Ω_{HF} denotes the outer 20% of frequency bins and the denominator represents the total spectral energy.

Color Statistics (6 dimensions). Per-channel mean and standard deviation across the R, G, and B channels, capturing color distribution properties and saturation uniformity.

2.5.3. Statistical Validation

Before training, we validate each feature's discriminative power using **two-sample t-tests** (real vs. AI-generated) and **Cohen's** d effect sizes:

$$d_i = \frac{\mu_i^{\text{real}} - \mu_i^{\text{AI}}}{\sqrt{(\sigma_i^{\text{real}\,2} + \sigma_i^{\text{AI}\,2})/2}}. \tag{10}$$

Features with non-significant p-values are optionally filtered. The remaining features are ranked by $|d_i|$; gradient entropy and high-frequency energy consistently emerge as the most discriminative.

2.5.4. Feature-Aware Weighting

Not all features contribute equally to classification. We propose **feature-aware weighting**: each feature dimension is multiplied by a scalar weight w_i derived from its normalized Cohen's d value before being fed into the classifier:

$$w_i = \frac{|d_i|}{\max_j |d_j|}, \quad \tilde{\mathbf{x}} = \mathbf{w} \odot \mathbf{x}, \tag{11}$$

where $\odot$ denotes elementwise multiplication. This soft gating directs the model's capacity toward the most separable features and provides an interpretable, data-driven feature selection mechanism without requiring a separate feature selection step.

2.5.5. MLP Classifier

The weighted feature vector $\tilde{\mathbf{x}} \in \mathbb{R}^{11}$ is passed through a lightweight multi-layer perceptron with architecture $11 \to 32 \to 16 \to 1$:

$$\hat{y} = \sigma\big(W_3\,\text{ReLU}(W_2\,\text{BN}(\text{ReLU}(W_1\tilde{\mathbf{x}})))\big), \tag{12}$$

where BN denotes batch normalization applied after the first linear layer. The network is trained with binary cross-entropy loss and the Adam optimizer.

2.6. Evaluation Protocol

All models are evaluated on the same held-out test split (80/20 stratified split). We report **Accuracy**, **ROC-AUC**, **F1 score** (macro and per-class), and **Average Precision (AP)** on both the full test set and the failure-case subset. Reporting results on both splits allows us to distinguish overall performance from robustness on genuinely difficult samples, and to measure the degree to which each approach degrades under distribution shift toward hard cases.

3. Results and Analysis

3.1. Statistical Entropy-Based Forensics

The proposed method demonstrates strong performance in distinguishing real and AI-generated images. On the test set, it achieves an overall accuracy of 92.35%, a ROC-AUC of 0.9086, and an average precision of 0.9717. Class-wise F1 scores indicate particularly high performance for detecting AI-generated images (F1 = 0.9552) while maintaining good discrimination for real images (F1 = 0.6345). These results show that the compact statistical feature representation effectively captures discriminative cues across multiple feature families, enabling reliable AI image detection.

3.2. Multimodal Vision-Language Classification

We present a systematic evaluation of the four multimodal configurations described in Section 2.3: (a) fine-tuned ViT, (b) zero-shot multi-prompt CLIP, (c) Encoder+CLIP with trained projection head, and (d) the logit-averaging ensemble of (a) and (c). Results are reported on both the full held-out test set and the Combined Failure Cases Dataset, allowing us to separately assess overall accuracy and robustness under distributional stress.

3.2.1. Overall Performance on the Test Set

Table 1 summarizes the per-model performance on the full test set ($n = 4{,}327$).

The **fine-tuned ViT** and **Encoder+CLIP** achieve nearly identical accuracy (95.84% vs. 95.82%), with the ensemble marginally improving to 96.00%. All three supervised models substantially outperform zero-shot CLIP (72.54%), confirming that visual artifact training is essential for real vs fake detection.

The **fine-tuned ViT** achieves strong per-class recall (0.98 Fake, 0.87 Real) with a macro-averaged F1 of 0.93, correctly classifying 3,486 fake and 661 real images. However, as discussed in Section 3.3, this comes at the cost of overfitting to dataset-specific artifacts, with disproportionate degradation on hard out-of-distribution cases.

Zero-shot CLIP performs substantially worse than the supervised models (72.54%, AUC 0.6721). While Fake-class F1 remains acceptable at 0.83, Real-class F1 collapses to 0.33, revealing that CLIP's semantic space does not encode the low-level distributional cues needed to distinguish human artwork from AI-generated imagery. Multi-prompt averaging reduces prompt sensitivity but does not resolve this representational gap.

Encoder+CLIP matches the fine-tuned ViT on accuracy (95.82%) and AUC (0.9800) while improving Real-class precision slightly (0.90 vs. 0.88), suggesting that grounding visual features in CLIP's semantic space provides a modest but consistent benefit on the harder real-image class.

Ensemble achieves the best overall accuracy (96.00%) and AUC (0.9824). The gain over individual models is modest in absolute terms but reflects genuine error complementarity: when the ViT and Encoder+CLIP disagree, the higher-confidence model's logits dominate the average, reducing both error types relative to either model alone. Confidence behavior is analyzed further in Section 3.4.

Table 1. Performance of Different Methods on the Full Test Set ($n = 4{,}327$), with Per-Class F1 Scores for Fake and Real Images

Model	Accuracy	ROC-AUC	F1 (Fake)	F1 (Real)
ViT (Fine-tuned)	0.9584	0.9824	0.97	0.88
Zero-Shot CLIP	0.7254	0.6721	0.83	0.33
Encoder + CLIP	0.9582	0.9800	0.97	0.88
Ensemble (Ours)	**0.9600**	**0.9824**	**0.98**	0.88
Statistical Entropy-Based Forensic (Ours)	0.9235	0.9086	0.9552	0.6345
PCA SVM (Baseline)	0.8530	0.8132	0.9114	0.5691
Frequency SVM (Ours)	0.9129	0.9099	0.9471	0.0.7525
Decision Tree Adaboost (Ours)	0.8787	0.8160	0.9254	0.6745

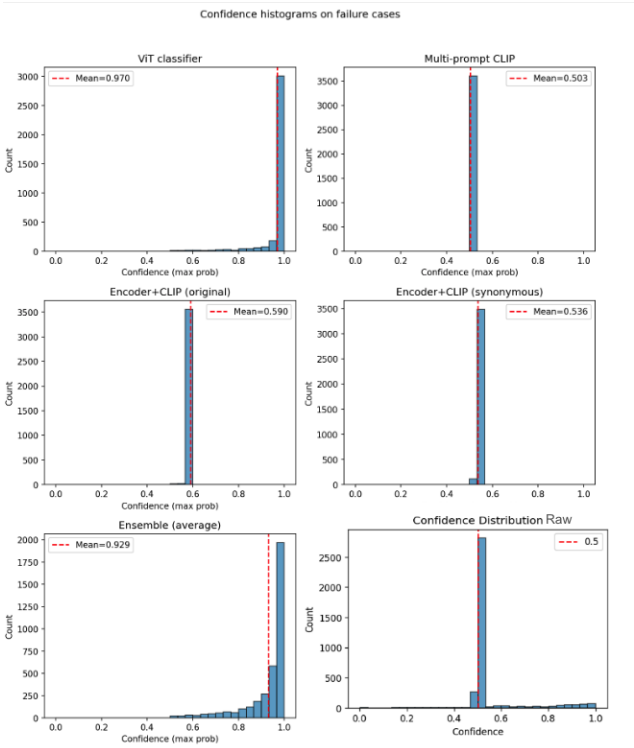


Figure 2. Confidence score distributions (maximum predicted probability) across all models on the Combined Failure Cases Dataset.

3.3. Performance on the Failure Cases Dataset

Table 2 reports performance on the Combined Failure Cases Dataset ($n = 3{,}610$), which contains only images misclassified by at least one model. This benchmark isolates the regime where detectors are most challenged and most likely to diverge in behavior.

3.3.1. Universal Performance Degradation

All models degrade on the failure cases, confirming that the benchmark successfully isolates genuinely difficult samples. The supervised models (ViT, Encoder+CLIP, Ensemble) drop by approximately 7 percentage points in accuracy, while zero-shot CLIP drops by nearly 10 points. The AUC drops are similarly structured: supervised models lose approximately 6 points while zero-shot CLIP loses 9 points. This pattern indicates that semantic-only models are disproportionately sensitive to hard cases–precisely the cases where low-level artifact evidence is weak or ambiguous.

3.3.2. Error Type Asymmetry

A striking pattern emerges in the error type breakdown. Among the 3,611 failure cases, 2,578 (71.4%) are Fake→Real misclassifications–AI-generated images incorrectly predicted as real–versus 1,033 (28.6%) Real→Fake errors. This 2.5:1 ratio reveals a systematic bias: across all methods, models are far more likely to be fooled by AI-generated images that have been engineered or selected to appear natural than to falsely flag authentic human artwork as synthetic.

This asymmetry has different implications across model families. The supervised ViT and ensemble models are biased toward Fake predictions (high fake recall, lower real recall), meaning their failures are predominantly Fake→Real: when they do err, they fail to catch AI images that have effectively evaded artifact detection. Conversely, the frequency-domain spectral classifiers (discussed in Section 3.5) exhibit the *opposite* bias, producing more Real→Fake errors. This complementary failure structure motivates cross-track ensemble strategies for future work.

3.3.3. Confidence Concentration at the Decision Boundary

Perhaps the most diagnostically significant finding from the failure cases is the distribution of model confidence scores. As shown in Figure 2, the vast majority of failure cases–across all models–yield predicted probabilities tightly concentrated around 0.5. This is not a gradual distribution but a sharp spike: models are not confidently wrong, they are *uncertain*. The failure cases are not adversarial in the classical sense, but they occupy a decision-boundary region where no evaluated model has learned reliable discriminative features.

Table 2. Performance of different models on the Failure Cases Dataset (n = 3,610).

Model	Acc.	AUC	F1 (Fake)	F1 (Real)	ΔAcc.	ΔAUC
ViT (Fine-tuned)	0.8892	0.9196	0.92	0.82	−0.0692	−0.0628
Zero-Shot CLIP	0.6263	0.5816	0.74	0.36	−0.0991	−0.0905
Encoder + CLIP	0.8856	0.9209	0.92	0.81	−0.0726	−0.0591
Ensemble (Ours)	**0.8898**	0.9197	**0.92**	**0.82**	−0.0702	−0.0627
Statistical Entropy-Based Forensic (Ours)	0.833	0.875	0.885	0.57	-0.1005	-0.0336
PCA SVM (Baseline)	0.8037	0.7932	0.8716	0.5837	−0.0493	−0.0179
Frequency SVM (Ours)	0.8792	0.8937	0.9181	0.7702	−0.0337	−0.0162
Decision Tree Adaboost (Ours)	0.8453	0.8332	0.8947	0.7089	−0.0334	−0.0172

This confidence concentration has a practical implication: a confidence-based abstention mechanism (refusing to classify images with predicted probability in $[0.45, 0.55]$) would correctly identify the majority of hard cases as uncertain, allowing downstream human review of exactly the samples where automated detection is least reliable.

3.4. Model Confidence Analysis

Table 3. Mean confidence (max predicted probability) on the failure cases dataset.

Model	Mean Confidence
ViT Classifier	0.970
Encoder+CLIP (original prompts)	0.590
Encoder+CLIP (synonymous prompts)	0.536
Multi-Prompt CLIP (Zero-Shot)	0.503
Ensemble (Ours)	0.929
Statistical Entropy-Based Forensic	0.865
Frequency SVM	0.929
Decision Tree Adaboost	0.881

3.4.1. ViT Classifier: High Confidence, Potential Overfit

The fine-tuned ViT produces a confidence distribution concentrated near 1.0 (mean: 0.970). While this yields high overall accuracy, the model fails confidently on hard cases, making confidence-based abstention ineffective and suggesting overfitting to dataset-specific artifacts. (Fig 2)

3.4.2. CLIP-Based Models: Low Confidence, High Uncertainty

Zero-shot CLIP produces a distribution centered at 0.5 (mean: 0.503), reflecting near-random uncertainty on failure cases. Encoder+CLIP variants show intermediate confidence (0.590 and 0.536), indicating partial calibration from projection head training. The low confidence of CLIP-based models might suggest they "know what they do not know," but their lower overall accuracy reveals the uncertainty stems from representational insufficiency rather than well-calibrated epistemic humility. (Fig 2)

3.4.3. Ensemble: Best Confidence Profile

The ensemble achieves the highest mean confidence (0.929) while producing a broader distribution with meaningful mass in $[0.7, 0.9]$-confident when both component models agree, appropriately uncertain when they diverge. This is the most desirable profile among all evaluated configurations. (Fig 2)

3.5. Frequency Domain Spectral Analysis

3.5.1. Overall SVM Performance and Hyperparameter Tuning

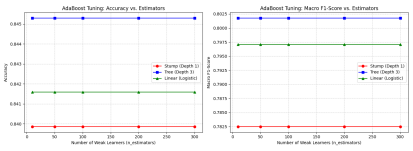
We first evaluate the performance of the Support Vector Machine (SVM) equipped with a Radial Basis Function (RBF) kernel on our 1D azimuthal profiles. The initial SVM achieved an overall accuracy of 85.6%.

While our SVM excelled at catching synthetic images (98% recall), it misclassified a significant portion of Real images (68% recall). When analyzing the failure cases, we observe that certain real images exhibit natural, high-frequency noise (e.g. sharp edges) that the RBF kernel tends to identify as a synthetic artifact. Thus, while the spectral domain is highly helpful for classification, natural variance in real-world images is a statistically non-negligible consideration for AI image detection.

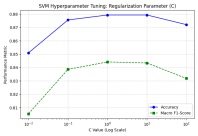
During hyperparameter optimization, we investigated the effect of the regularization parameter C to balance margin maximization and training error. We observed that performance (Accuracy and Macro F1) steadily increased as C was scaled from 0.01 to 1.0, peaking at an optimal accuracy of 0.8792. However, at a highly penalized threshold ($C = 100$), accuracy degraded to 0.8720. This confirms that an overly rigid margin forces the model to overfit to spectral outliers, whereas $C = 1.0$ provides the optimal mathematical balance.

3.5.2. Feature Extraction: Azimuthal Integration vs. PCA

To rigorously evaluate the efficacy of our engineered features, we compared our Azimuthal Integration against a baseline of Principal Component Analysis (PCA) applied



(a) AdaBoost Parameter and Architectural Tuning. Accuracy (left) and Macro F1-Score (right) indicate rapid convergence within the first 10 estimators, with Depth-3 trees outperforming standard stumps.



(b) SVM Hyperparameter Tuning. The optimal balance between margin maximization and error tolerance is achieved at $C = 1.0$.

Figure 3. Hyperparameter tuning results for classical classifiers evaluated on 1D azimuthal profiles.

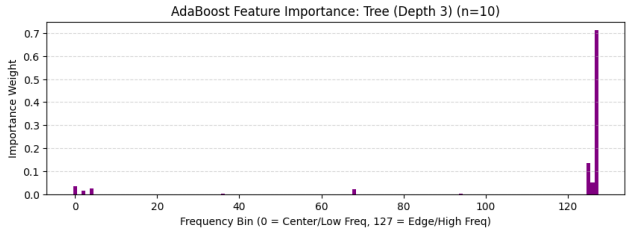


Figure 4. AdaBoost feature importance (Depth-3 Tree, $n = 10$). The ensemble correctly learns to prioritize the extreme high-frequency bins, perfectly aligning with the divergence observed in the radial profiles.

directly to the flattened 2D Fourier spectrum. Table 2 summarizes this comparison.

Despite capturing 95% of the dataset's total variance, the PCA-based SVM achieved an accuracy of only 80.4%, which is much weaker than the Azimuthal Integration model (87.9%).

Thus, maximizing global variance does not necessarily equate to maximizing class separability. Deepfake fingerprints account for only a small fraction of the total statistical variance, leading traditional dimensionality reduction methods like PCA to dilute discriminative ability.

3.5.3. AdaBoost: Base Estimators and Ensemble Tuning

We further evaluated classical ensemble methods by training AdaBoost models on the 1D azimuthal profiles. Table 4 highlights the performance hierarchy based on the mathematical expressivity of different base estimators.

Table 4. AdaBoost performance across different weak learner architectures.

Weak Learner	Accuracy	Macro F1
Decision Stump (Depth 1)	∼0.840	∼0.783
Linear (Logistic)	∼0.842	∼0.797
Decision Tree (Depth 3)	∼**0.845**	∼**0.802**

Decision Stumps yielded the lowest performance. Because a stump partitions the space using a single frequency bin per iteration, it struggles to capture the continuous structural violation of the power-law decay. Linear (Logistic) Models achieved moderate improvements by computing a weighted sum across all 128 bins simultaneously, accurately modeling the global slope of the spectral decay. However, as shown by Figure 4, Depth-3 Decision Trees achieved the highest overall metrics, thus capturing cross-bin conditional relationships most robustly classified images.

A notable observation from our parameter tuning curves is the stagnation of test performance; for all three architectures, the accuracy and Macro F1-scores remained perfectly flat from 10 estimators up to 300. Because our azimuthal integration aggressively reduces the original 256×256 spatial domain into a highly concentrated 128-dimensional spectral vector, the discriminative "GAN fingerprints" are highly pronounced. The AdaBoost algorithm successfully isolates and leverages these dominant high-frequency signals within the first 10 boosting iterations for fast convergence.

4. Conclusion

We compared three AI-generated art detection paradigms: vision-language classification, frequency-domain spectral analysis, and statistical entropy-based forensics. Across all three, a consistent theme emerged each method captures distinct artifact signatures, and their complementary strengths point toward a unified architecture. In the multimodal stream, supervised fine-tuning proved decisive: a fine-tuned ViT (95.84%, AUC 0.9824) substantially outperformed zero-shot CLIP (72.54%, AUC 0.6721), with multimodal grounding adding a modest further gain. A logit-averaging ensemble achieved the best overall accuracy at 96.00%, while failure cases clustering near p ≈ 0.5 suggest confidence-based abstention as a promising refinement.

Frequency-domain analysis revealed that physics-informed azimuthal integration (87.9% SVM accuracy) significantly outperformed a PCA variance baseline (80.4%), confirming that radial constraints better isolate generative artifacts. Depth-3 AdaBoost ensembles further improved over single stumps, indicating these artifacts span multiple high-frequency bands. However, spectral classifiers remain susceptible to natural high-frequency noise in authentic artwork.

The entropy-based forensic approach demonstrated that handcrafted statistical features capturing entropy, sharpness, frequency, and color distribution can achieve strong detection without any deep neural networks, confirming that synthetic images exhibit measurable deviations from natural image statistics.

Overall, these results suggest that combining multimodal, spectral, and statistical cues may provide a promising direction for AI-generated image detection.

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